

JEE MAIN REVISION MASTER: PROPERTIES OF TRIANGLES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Sine Rule, Cosine Rule & Triangle Geometry	4 PYQs	Triangle Properties

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Sine Rule, Cosine Rule & Triangle Geometry	$\frac{a}{\sin A} = 2R$, $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, $\Delta = rs = \frac{abc}{4R}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Sine Rule, Cosine Rule & Triangle Geometry (Moderate Yield — 4 PYQs)

Core Principle: Sine rule: $(a / \sin A) = (b / \sin B) = (c / \sin C) = 2R$. Cosine rule: $\cos A = (b^2 + c^2 - a^2) / 2bc$. Projection formulas: $a = b \cos C + c \cos B$. Area $\Delta = (1 / 2)ab \sin C = \sqrt{s(s-a)(s-b)(s-c)} = rs = (abc / 4R)$.

■ **Caution / JEE Pitfall:** In an equilateral triangle: $R = 2r$ (circumradius is twice the inradius).

■ **Speed Trick:** In right-angled triangle: Inradius $r = (a+b-c) / 2$ where c is the hypotenuse.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The angles of a triangle are in A.P and the greatest angle is double the least angle, then sine of the third angle is..." [Ans: A]
- [2024 Shift] Q#2 (MCQ): "If $\triangle ABC$ is right angled at C , then the value of $\tan A + \tan B$ is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The area of a triangle with vertices $(-3,0)$, $(3,0)$ and $(0,k)$ is 9 sq units, find the value of k is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The angles of a triangle are in A.P and the greatest angle is double the least angle, then sine of the third angle is

- (A) $\frac{\sqrt{3}}{2}$ **[Correct]**
- (B) $\frac{1}{2}$
- (C) $\frac{1}{\sqrt{2}}$
- (D) 0

Correct Answer: A

Solution Analysis: Let the three angles of the triangle be in A.P. So we can take them as $a-d, a, a+d$. Since the sum of angles of a triangle is 180° , $(a-d) + a + (a+d) = 180^\circ$ $3a = 180^\circ$ $a = 60^\circ$. So the middle angle is 60° . Now, the greatest angle is double the least angle. Greatest angle $= a+d$ Least angle $= a-d$. So, $a+d = 2(a-d)$. Putting $a = 60^\circ$, $60^\circ + d = 2(60^\circ - d)$ $60^\circ + d = 120^\circ - 2d$ $3d = 60^\circ$ $d = 20^\circ$. Thus the angles are $40^\circ, 60^\circ, 80^\circ$. The third angle in the A.P. is the middle angle, 60° . So, $\sin 60^\circ = \frac{\sqrt{3}}{2}$.